§6 Minimal structure of ACS relations — part 1 of 4

Reference: sec:rel_min, subsec:rel_objects, subsec:rel_unidirect

Source: paper/section6_relational.tex, lines 1–711 (opening and scope; the set of external objects and the three-way classification; the local size indicator, the finite observation window and the unidirectional relation descriptor), at commit e6e12cd. Status: draft. §6 has four parts: part 2 (06_relational_2.md, 712–1331: bidirectional composition, relation quality and its evaluable domain, formalization of Q), part 3 (06_relational_3.md, 1332–1791: sign classification of Q through the relation to T-STAB), part 4 (06_relational_4.md, 1792–2171: representative categories and six families; scope, hand-off and summary). §5 is in 05_dynamics_1.md and 05_dynamics_2.md.

Opening and scope

Reference: sec:rel_min, rem:rel_scope, rem:rel_to_sec7_intro

§6 fixes the minimal structure of a relation that can hold between a single ACS Σ_i and an external object Y_j, in the order: object classification, unidirectional relation description, bidirectional composition, relation quality, representative categories. Relations are not sorted into ethical good and bad; the framework describes changes in feasibility-related quantities that can be recorded and evaluated on the individual side while a relation continues. A record includes classification-C theoretical description and does not require the ACS itself to observe or compute each value on the same scale. $d_{\mathrm{eff},i,t}$ and $\hat{\tau}_{i,t}$ of §5 are not redefined but connected to finite-window differences attached to relation records; the local size indicator sz_i and the difference notation Δ_L do not change the eight components of Desc_t. The differences do not express the sole causal effect of an external object and do not replace the conditions for blockage and collapse of §2. Unidirectional versus bidirectional is a difference in the level of description, not in the value of the object or the merit of the outcome of a relation; the same label can be accompanied by different changes depending on cost, environmental conditions and the state of the ACS, and a category name is not a rule that fixes the sign or cause of the change in advance.

Scope of this section (`rem:rel_scope`). The section covers the three-way classification of external objects, unidirectional relation descriptors, bidirectional composition between ACSs, relation quality, and the 18 representative categories with their six families. A finite-window difference of a unidirectional descriptor is defined only when the required time domain and record conditions hold; totality (six components obtainable at any object and time whenever a relation exists) is not required. Classifiability of an external object, decidability of a relation category and well-formedness of a finite-window difference are separate matters. Conditions for establishing categories, thresholds, strength, persistence, releasability, substitutability and the correspondence to recognition are not fixed. Deterministic auxiliary refinement goes to Phase 1.α, and probabilistic establishment, transition and termination, continuous time and social-network dynamics to Phase 2 (L8 of `subsec:limitations_list`, `subsec:phase1a`). The identification and candidate formation of §4 and the judgments of blockage and collapse of §2 and §7 are not redefined. That a classifiable object has no determined category or difference does not by itself impair well-formedness of the section. The temporal development of relations and the generating rule of relation networks are not included in the descriptors.

Relation to the blockage and collapse section (`rem:rel_to_sec7_intro`). §7 treats blockage $A_{i,t} = \emptyset$ and collapse $\mathrm{Col}_{i,t} = 1$ under separate conditions. Differences attached to relation records can be descriptive material for analyzing the blockage path set $\mathfrak{P}_{\mathrm{blk}}$ and the collapse-reaching path set $\mathfrak{P}_{\mathrm{col}}$, but the sign of a difference, a relation category or the existence of a descriptor alone does not decide membership in those path sets. Ordinary ACS state quantities after collapse are not required for computing differences. When relation records accompany a path description, membership is judged by the state and trajectory conditions of the later section. The absence of a record is not equivalent to the absence of a relation.

External objects and the classification map

Reference: subsec:rel_objects

Definition (def:external_object_set): set of external objects. For a fixed individual ACS Σ_i, $\mathcal{Y}_{-i}$ is the set of external objects, excluding itself, adopted as objects of relation; individual objects are $Y_j \in \mathcal{Y}_{-i}$. $(-i)$ is a marker for "excluding itself" and not an exponent. The set is a domain of objects to be described and assumes neither an internal state space, decision mechanism or self-maintenance mechanism of each Y_j nor reliable observability over time. Non-emptiness of $\mathcal{Y}_{-i}$ and the holding of a relation with every object are not required. Belonging to the object set is distinguished from the six-component descriptor with that object being defined at a given time. Words such as observation, reference, action, dependence and support show modes in which an object can be related and give no conditions for their holding. The index j is a marker for tracking the same object in relation records and adds no ACS structure to the object.

Role of the set of external objects (rem:external_object_role). $\mathcal{Y}_{-i}$ is an object domain different from the state space X_i of Σ_i; an external object Y_j is also a different kind of object from the external environment state $e_t \in E_{\text{ext}}$, and is neither identified with e_t nor read as $Y_j \in E_{\text{ext}}$. A non-ACS object is not automatically assigned X_j, $A_{j,t}$ or ordinary ACS state quantities. The set E of required environmental factors and $\mathcal{Y}_{-i}$ are different objects of description, and a one-to-one correspondence of some Y_j with a particular required factor is not assumed. Whether Y_j corresponds to a slot $a \in \mathcal{A}_O$ of the action-response component, to a background component, or to an internal object token k of §4 is given by the describer, and this paper does not define the correspondence. The descriptors use only values on the side of Σ_i, so they are defined without this correspondence. The section does not develop the internal dynamics of external objects, and the same object can be an object of different relation records from different ACSs; changing the viewpoint of a record does not give a state equation of the object side.

Identification of external objects versus action candidates (rem:external_object_cand). Internal identification concerning attribution of an external object to the self–environment boundary connects to the identification state of M1-b (def:id_vector). The value range $\{\mathsf{self}, \mathsf{env}, \mathsf{und}\}$ of the boundary-attribution axis does not mean the identity of each Y_j is settled internally at every time. type_i of §6 is a classification map for describing relations; it is neither the identification state of M1-b nor an assumption that the ACS internally holds the same classification. The candidate set $\text{Cand}_t \subseteq U$ of M3 (def:m3_candidate_set) is a set of action candidates, not of external objects, and the fit evaluation fit_t (def:m3_fit) differs from object identification (rem:fit_vs_identity). This paper does not state $Y_j \in \text{Cand}_t$ from $Y_j \in \mathcal{Y}_{-i}$ as a general consequence. Identifying an object, the object's fitting the request at the time, and an action toward the object being generated as a candidate are different conditions, and the classification label $\text{type}_i(Y_j)$ alone does not give these mechanism-level judgments. A correspondence in records presupposed in this section is distinguished from the mechanism-level judgment of whether M1-b actually identified the object.

Definition (def:object_type): three-way classification of external objects. A total single-valued descriptive classification map relative to the fixed Σ_i is $\text{type}_i : \mathcal{Y}_{-i} \to \{\text{ACS}, \text{agent}, \text{obj}\}$, with preimages $\mathcal{Y}_{\text{ACS},-i}, \mathcal{Y}_{\text{agent},-i}, \mathcal{Y}_{\text{obj},-i}$. The three values are descriptive labels for an external object treated as an ACS, an agent not treated as an ACS, and a non-agent object. The map fixes neither the normative status of an object nor the accuracy of actual internal recognition. Any preimage may be empty. That an actual object may hold several properties is not inconsistent with assigning a single label in description.

Assumption (ass:obj_stable): fixed classification of external objects. While the same object is tracked in this section, type_i is not changed by time t. It does not prohibit actual changes in an object's attributes and asks only for temporal fixing of the classification used. Introduction status: S (structural, restriction of the description domain). Strength: weak. Relied on in subsec:rel_objects and subsec:rel_bidirect.

Meaning of the fixed-classification assumption (`rem:obj_classification`). The assumption restricts the classification during the adopted description period and is not a claim that physical properties, agency or ACS structure of the object are actually invariant. Treating an object whose classification changes midway needs a separate time-dependent classification, which the section does not give (for example, switching from "non-agent object" to "agent").

Exclusivity of the three classes (`rem:object_partition`). Because type_i is a total single-valued map, preimages of different values are disjoint and their union is $\mathcal{Y}_{-i}$: $\mathcal{Y}_{\mathrm{ACS},-i} \dot\cup \mathcal{Y}_{\mathrm{agent},-i} \dot\cup \mathcal{Y}_{\mathrm{obj},-i} = \mathcal{Y}_{-i}$. This is a consequence of the definition and not a new independence assumption. Non-emptiness of each preimage does not follow, and the equality holds also when no external object exists.

Position of the three classes (`rem:object_partition_role`). Exclusivity is that of the values of the map and does not deny that an actual object has several properties at once. The classification is relative to Σ_i, and another ACS Σ_k may adopt a different classification map. The undetermined category `und` of M1-b is an internal identification value and does not add a fourth external-object type. Describing $\mathrm{type}_i(Y_j) = \mathrm{ACS}$ does not mean that the ACS itself has recognized the object as `env` or that another ACS assigns the same type.

Object type, blockage and collapse (`rem:objects_to_sec7`). This paper does not state $A_{i,t} = \emptyset$ or $\mathrm{Col}_{i,t} = 1$ from the type of an external object alone as a general consequence. $\mathcal{Y}_{-i}$ and type_i only fix the object domain of relation description; membership in blockage or collapse-reaching paths uses the conditions of §7 separately.

Unidirectional relation descriptors and the finite observation window

`Reference: subsec:rel_unidirect`

Definition (`def:rel_size_functional`): local size indicator of the feasible action set (classification C). With the action space U_i of the fixed Σ_i, a non-empty family $\emptyset \neq \mathcal{D}_{\mathrm{sz},i} \subseteq \mathcal{P}(U_i)$ to be evaluated and a non-negative real-valued descriptive map $\mathrm{sz}_i : \mathcal{D}_{\mathrm{sz},i} \to \mathbb{R}_{\geq 0}$ may be placed; sz_i is called a *local size indicator*. "Size" is a descriptive name for the evaluated set; the type specification alone does not give monotonicity with respect to inclusion, non-degeneracy, agreement with cardinality for finite sets, or $\mathrm{sz}_i(\emptyset) = 0$. Differences in the indicator are therefore not read in general as increase or decrease of feasible actions, and $\mathrm{sz}_i(A) = 0 \Rightarrow A = \emptyset$ is not placed as a property. Whether to adopt the indicator, its domain of evaluation and its concrete form are not fixed in the general form; if $\emptyset$ is not in the domain, $\mathrm{sz}_i(\emptyset)$ itself is not required. For time differences of one individual, one domain and one map are used at both ends of the window; a difference of values with the map exchanged over time is not read as a time change of the same indicator. The indicator is not identified with $d_{\mathrm{eff},t}$ of §5, with the feasibility constraints of §3, or with the response diversity $\mu_{\mathrm{eff},t}$, and no ordering between sz_i and $\mu_{\mathrm{eff},t}$ is assumed. When the indicator is actually used, the feasible action sets evaluated are confirmed to belong to $\mathcal{D}_{\mathrm{sz},i}$. Since no concretization is given, additional comparisons or inferences needing non-trivial size properties are not stated as general consequences; the sign of $\Delta_L \mathrm{sz}_i(A_{i,t})$ is read as the sign of a numerical difference under the same indicator. Cardinality for finite sets and fixed-precision covering numbers for general sets are candidates for future concretization but are not adopted as the definition of sz_i.

Definition (`def:finite_window_diff`): difference notation on a finite observation window (classification C). Take $L \in \mathbb{N}_{\geq 1}$ and $t \in \mathbb{N}_0$. While one relation record or bidirectional composition is evaluated, the window length L is fixed in common; records with different window lengths are not mixed as the same $\mathrm{Rel}_{i\leftarrow j,t}$. For $t \geq L$ the finite observation window is $W_{L,t} = \{t-L, \dots, t\}$. For a time series $z_{i,s}$ of one real-valued quantity of Σ_i defined at both ends $s = t-L, t$, the backward difference is $\Delta_L z_{i,t} = z_{i,t} - z_{i,t-L}$; for $t < L$ or when either value is missing, it is not defined. The definition of the difference itself does not require the values at the intermediate times. The three quantities used are $\mathrm{sz}_i(A_{i,s})$, $d_{\mathrm{eff},i,s}$ and $\hat\tau_{i,s}$. For the first, $A_{i,t}, A_{i,t-L} \in \mathcal{D}_{\mathrm{sz},i}$ is required and the same sz_i is used at both ends. The latter two are the quantities of §5 written per individual following the convention on levels of description of §2; differences are taken only where both end values are defined, and since $\hat\tau_{i,s}$ uses records from s to $s+\tau$ (`def:s5_prediction_coefficient`), its difference is taken only at times where the value is defined. Real-valuedness of each difference is a consequence under this domain of application and not a general claim of measurability or internal computability. To incorporate

a difference into a unidirectional descriptor as a *change accompanying the relation record* with an external object Y_j, it is further required that the relation continue at all times in the window and that the index j refer to the same object. When the ACS-record valid time domain of §7 is used for a trajectory record σ of the same individual Σ_i, $W_{L,t} \subseteq I_{\mathrm{ACS}}(\sigma)$ is required (`def:acs_valid_time_domain`); this does not mean the three quantities can actually be obtained at all times. An undefined value is not replaced by zero in a window failing the condition. At $t = L$ the left end is time 0 and no negative-time history is referred to. Even if both end differences are computable, without confirmation of relation continuity and object identification in the window the difference is not incorporated into the record of the relation with that object.

Limits of finite-window differences and delegation of the size functional (`rem:finite_window_diff_limit`). Δ_L is a backward difference of finitely many discrete times, not a continuous-time derivative or a difference from the case where Y_j did not exist. The window $W_{L,t}$ contains $L + 1$ times. Outside $I_{\mathrm{ACS}}(\sigma)$ ordinary ACS state quantities are not required, and lost quantities are not interpolated over a window straddling the first time collapse holds. Blockage $A_{i,s} = \emptyset$ is not synonymous with collapse, so the inclusion of a blockage time in a window does not by itself exclude all differences. However, if blockage occurs at an end point and $A_{i,s} = \emptyset \notin \mathcal{D}_{\mathrm{sz},i}$, then $\mathrm{sz}_i(A_{i,s})$ is outside the domain and $\Delta_L \mathrm{sz}_i(A_{i,t})$ cannot be defined; in that case the six-component descriptor of `def:rel_unidirect` is not constructed even if the differences of $d_{\mathrm{eff},i,t}$ and $\hat{\tau}_{i,t}$ are each well-formed. Even when the empty set is in the evaluation domain, its indicator value need not be zero. Whether a difference is defined is judged by the values, the domain and the time domain of the adopted trajectory record. Even if the indicator is introduced, this paper does not state from a decrease of its value an inclusion relation of sets or $A_{i,t} = \emptyset$ as a general consequence. Concretization, finite approximation and fixed-precision covering numbers go to Phase 1.α (`subsec:phase1a`) and measure-theoretic refinement to Phase 2. This delegation is not an existence claim that a non-degenerate size indicator is automatically obtained on any compact action space.

Definition (`def:rel_unidirect`): unidirectional relation descriptor (classification C). Fix Σ_i, $Y_j \in \mathcal{Y}_{-i}$, a positive window length L and a time t; the window length is fixed over the recorded period. When the three differences of `def:finite_window_diff` are defined under the conditions for relation records, and a category $\mathrm{cat}_{i \leftarrow j,t} \in \mathcal{C}_{\rightarrow}$, a cost $\mathrm{cost}_{i \leftarrow j,t} \in \mathbb{R}_{\geq 0}$ and a perturbation $p_{\mathrm{pert},i \leftarrow j,t} \in P_{\mathrm{pert}}$ are recorded, the unidirectional relation descriptor from Y_j to Σ_i is the six-tuple $\mathrm{Rel}_{i \leftarrow j,t} = (\mathrm{cat}, \mathrm{cost}, p_{\mathrm{pert}}, \Delta_L \mathrm{sz}_i(A_{i,t}), \Delta_L d_{\mathrm{eff},i,t}, \Delta_L \hat{\tau}_{i,t})$. $\mathcal{C}_{\rightarrow}$ is the 15 labels of `def:category_set` and P_{pert} the non-empty perturbation range of `def:rel_unidirect_type`. The first three are records on relation category, non-negative cost and perturbation; the last three are differences of values on the side of Σ_i recorded and evaluated while the relation continues. If the last three are undefined, no six-component descriptor is placed even if the first three are recorded; conversely, if the three differences are defined but no category and the like are adopted, no six-component descriptor tied to a particular Y_j is constructed. The subscript $i \leftarrow j$ is a direction of description and j does not appear in the formulas of the last three, so the descriptor does not identify a unique causal contribution of Y_j; the same numerical differences can appear in several descriptors when relations with several objects are recorded for the same Σ_i and window, and this coincidence is not read as equal action or an allocation of contribution. The definition does not assign six components to objects and times where the conditions fail. The cost is a non-negative record of the resource load accompanying the relation, and cost = 0 does not mean the relation does not hold. The symbol p does not denote probability, and no category is judged here by the sign or magnitude of a perturbation value. Even with the same last three components, the category, cost and perturbation of the first three can differ.

Persistence of the object index and non-causality of differences (`rem:object_index_persistence`). The index j refers to the same external object over all times of $W_{L,t}$ and the relation with it continues; this is a condition for adopting a relation record. Object identification relates to the identification of M1-b (`def:id_vector`), but no judgment rule settling individual identity from the identification state is given. That the boundary-attribution axis shows `env` and that records at different times are judged to belong to the same individual Y_j are different conditions. The persistence of j is a correspondence requirement placed when adopting the record of the window and does not additionally assume the existence or accuracy of an

internal identification procedure of M1-b. The request-fit evaluation of M3 is a function separate from object identification (`rem:fit_vs_identity`). The three differences rest on values of Σ_i and the right side has no j; no rule subtracts contributions of other objects, actions or environmental conditions present at the same time. A difference is thus a "change recorded while the relation continued", not a counterfactual causal effect, a proof that a category holds, or an allocation of contributions. When relations with two objects continue in the same window, the same three differences can be listed in both descriptors; their values are not added to count the individual-side change twice, nor is each object read as having produced the whole difference.

Notational caution (`rem:rel_unidirect_notation`). The direction of $\mathrm{Rel}_{i\leftarrow j,t}$ shows the viewpoint of description and does not denote one-sided determination or domination. The p of $p_{\mathrm{pert},i\leftarrow j,t}$ abbreviates perturbation and is not probability; no distribution, expectation or probabilistic transition rule is assigned to it. The arrow is a direction of information description and does not require the object to have intention.

Definition (`def:rel_unidirect_type`): type of the unidirectional relation descriptor. At a time where the descriptor is defined, the types of the six components are $\mathrm{cat} \in \mathcal{C}_{\rightarrow}$, $\mathrm{cost} \in \mathbb{R}_{\geq 0}$, $p_{\mathrm{pert}} \in P_{\mathrm{pert}}$, and $\Delta_L \mathrm{sz}_i(A_{i,t}), \Delta_L d_{\mathrm{eff},i,t}, \Delta_L \hat{\tau}_{i,t} \in \mathbb{R}$. P_{pert} is a non-empty range expressing perturbing influence; its sign, order and concrete scale are not fixed. P_{pert} is not assumed to carry an order, nor are perturbation amounts assumed arithmetically additive with the cost or the three differences. Real-valuedness of the differences expresses well-formedness when the domain of differences and the ranges of the quantities of §5 hold. Finite label sets, ordered sets or real intervals can be considered as concretizations, but this paper adopts none. Comparison by perturbation size cannot in general be made without defining an additional type.

Role of type specification (`rem:rel_unidirect_type_role`). The type gives only the kind of each component and gives no measurement method, estimation error, category threshold, additivity or intensity scale of perturbation. This paper does not state, from a positive difference of sz_i, an expansion in the inclusion sense of the action set, nor from a negative difference, blockage, as a general consequence. The ACS interior is not required to compute and hold the whole classification-C descriptor in the same form.

Descriptor and feasibility-related quantities (`rem:rel_descriptor_viability`). $\mathrm{sz}_i(A_{i,t})$ is a local indicator adopted here, and $d_{\mathrm{eff},i,t}$ and $\hat{\tau}_{i,t}$ refer to the per-individual quantities of §5. Whether the three differences are positive, zero or negative, this paper does not state from the sign alone $A_{i,t} = \emptyset$, $\mathrm{Col}_{i,t} = 1$, recoverability, the truth of a time-indexed satisfaction predicate of the three principles, or a causal contribution of a particular object as a general consequence. These are quantities handled by different definitions and conditions.

Assumption (`ass:unidirect_scope`): object range of unidirectional description. For an object Y_j, a window length L and a time t at which a unidirectional descriptor is actually adopted, it is required that the category, cost and perturbation are recorded in the types of `def:rel_unidirect_type` and that the three differences satisfy `def:finite_window_diff`. The assumption does not require that a relation holds with any external object at any time, that six components can be recorded, or that the set of adopted tuples exists or is non-empty. It specifies well-formedness of adopted records conditionally and is not an existence assumption that the relation between external objects and ACS states is generally measurable. Introduction status: S (structural, well-formedness of adopted records). Strength: weak. Relied on in `subsec:rel_unidirect` and `subsec:rel_bidirect`.

Application to non-ACS objects (`rem:unidirect_nonacs`). Non-ACS agents and non-agent objects can also be objects of unidirectional description when the six components on the side of Σ_i satisfy the record conditions. X_j, $A_{j,t}$, $d_{\mathrm{eff},j,t}$ and the like of Y_j are not assumed. For objects and times where no record is adopted, no descriptor is made by filling unknown values with zero. Even for a non-agent object, it is formally possible to list individual-side differences where the object constrains the individual's choices or involves a resource load, but this is not evidence of decision-making or self-maintenance by the object itself.

Connection to the blockage and collapse section (`rem:to_section7`). Relation records can be input to the path descriptions of §7, but the existence of $\mathrm{Rel}_{i\leftarrow j,t}$ or the sign of a difference alone does not decide $\sigma \in \mathfrak{P}_{\mathrm{blk}}$ or $\sigma \in \mathfrak{P}_{\mathrm{col}}$. When relation records accompany the analysis of a path, the conditions for blockage and collapse along the trajectory follow the definitions of §2 (`def:blockage_sec2`, `def:collapse`). A category name of a relation is not a logical formula replacing those conditions.

Check status. This summary adds no new definition, assumption or result. Labels in backticks are shared with the manuscript. It has been checked against the manuscript by the drafting model and audited by an external AI model. It remains a draft until the author checks it against the manuscript. No independent human third-party verification is claimed.